

14.5
20

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Multimedia "ECE 359"

Q1 →

(a)

o

255

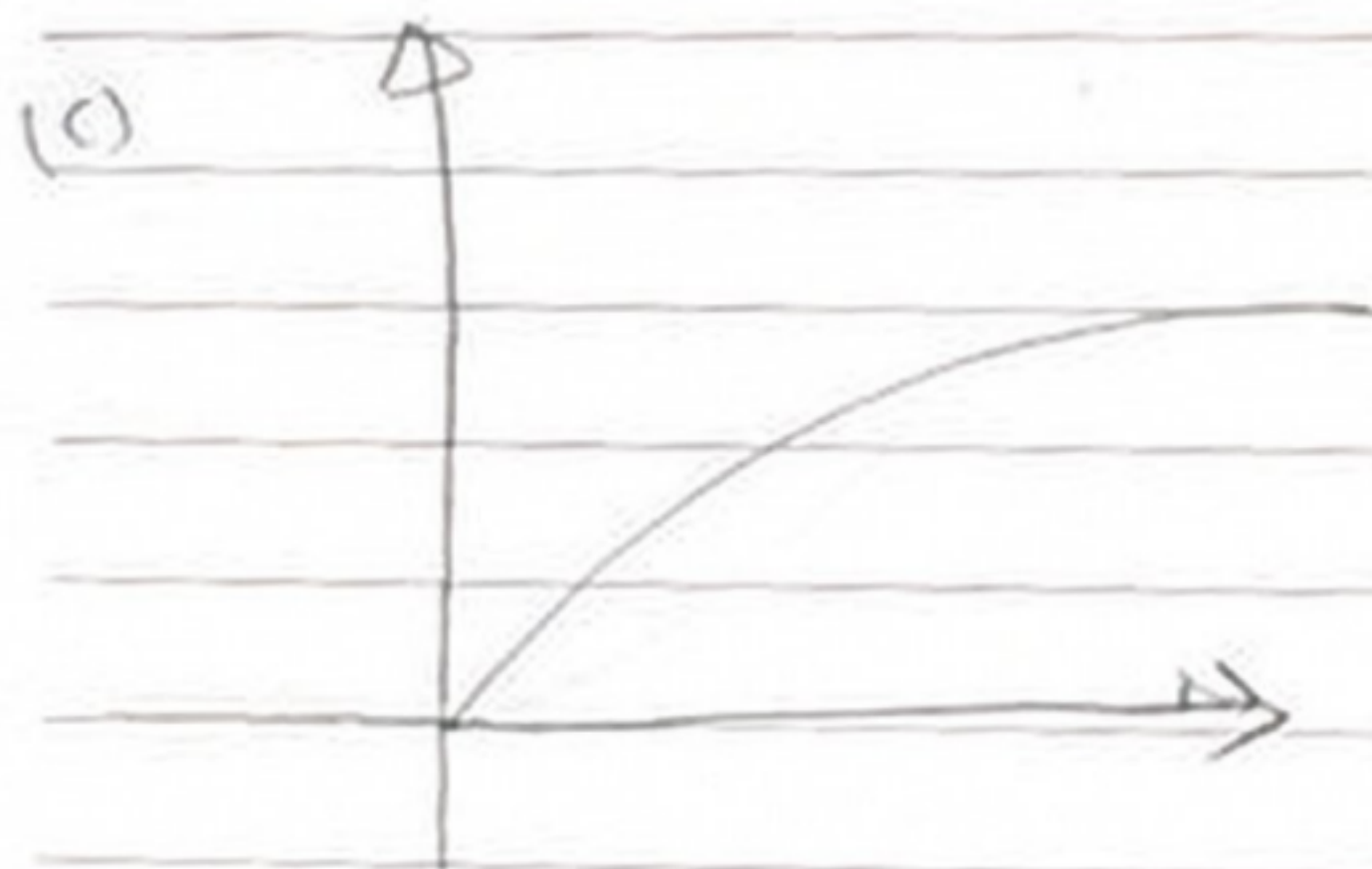
$$0 < r < \frac{2^K}{2} - 1$$

$$\frac{2^K}{2} < r < 2^K + 1$$

-0.5 unclear classification

(b) we want to increase darkness to make the unclear writing clear
 transform of 71 → darken image

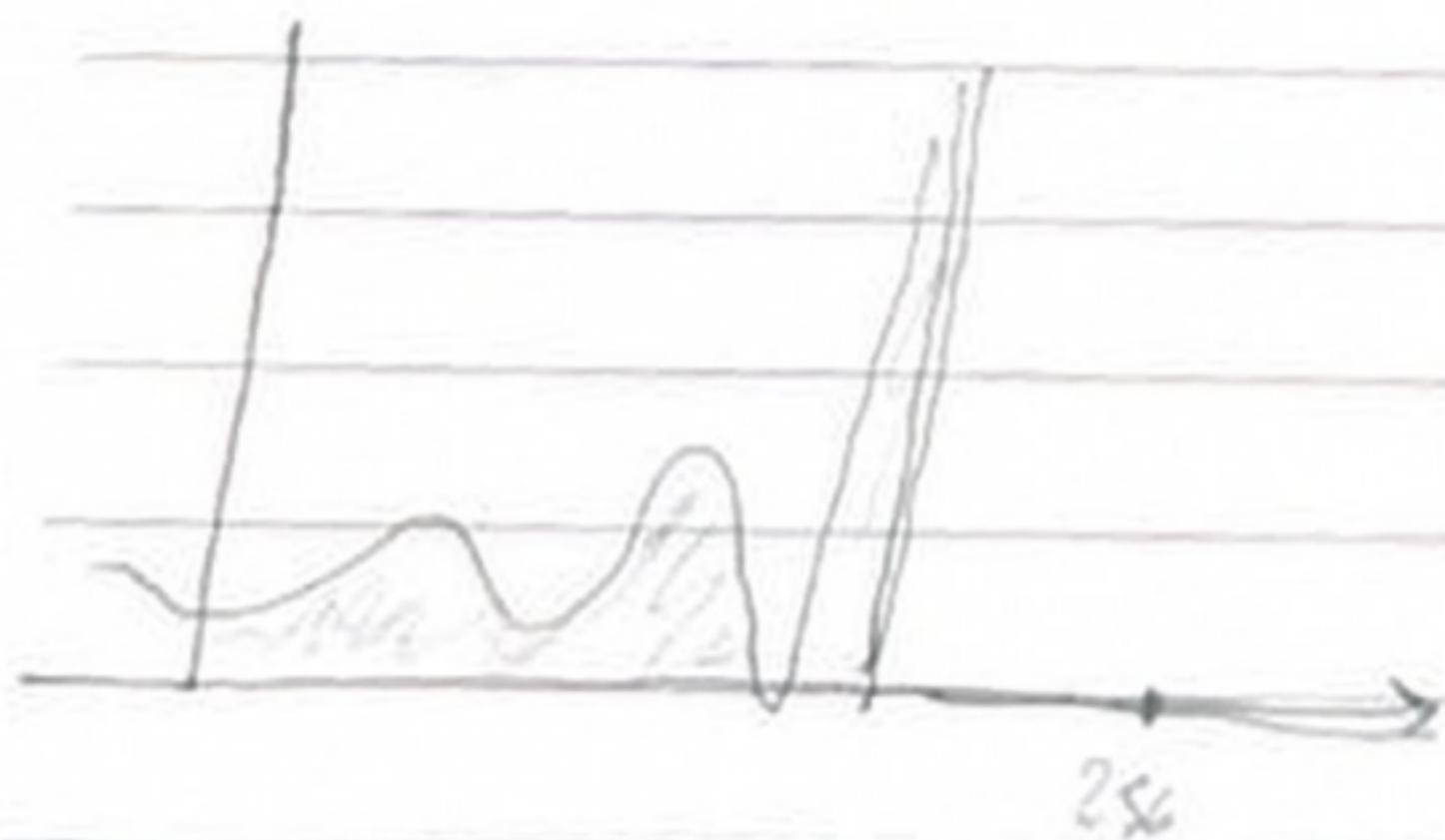
-0.5 not practical



Not the required
 we want shading to be white
 or disappear

(d) Shift Graph to Left

and image is not binarized (-0.5)



Q2 →

$$A \rightarrow (3, 3)$$

$$A' \rightarrow (7, 3)$$

$$S_x \xrightarrow{\begin{matrix} \times \\ \times \end{matrix}} \frac{7}{11} \begin{matrix} 2^{nd} \rightarrow 5.5, 1^{st} \rightarrow 3.5 \\ 1^{st} \rightarrow 4, 2^{nd} \rightarrow 3.2 \end{matrix}$$

$$S_y \rightarrow \frac{5}{4}$$

$$x' = S_x x + t_x$$

$$y' = S_y y + t_y$$

$$7 = \frac{7}{11} \times 3 + t_x \checkmark \quad \therefore t_x = \frac{56}{11}$$

$$3 = \frac{5}{4} \times 3 + t_y \checkmark \quad \therefore t_y = -\frac{3}{4}$$

$$T_{matrix} \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix}$$

Final answer incorrect -0.5

③

Q₃ →T_B → Negative (Bright ⇌ dark) → Graph E → H₃T_n → Quality is better → Graph D → H₂ not best answerT_C → darken the image → shifted left → H₄ → ~~Graph C~~ (-1/3)T_D → brighten image → shift right → H₁ → ~~Graph B~~ (-1/3)T_E → ~~Graph A~~ (-1/3) → H₅ → increasing darkness & brightness

→ This is identity graph

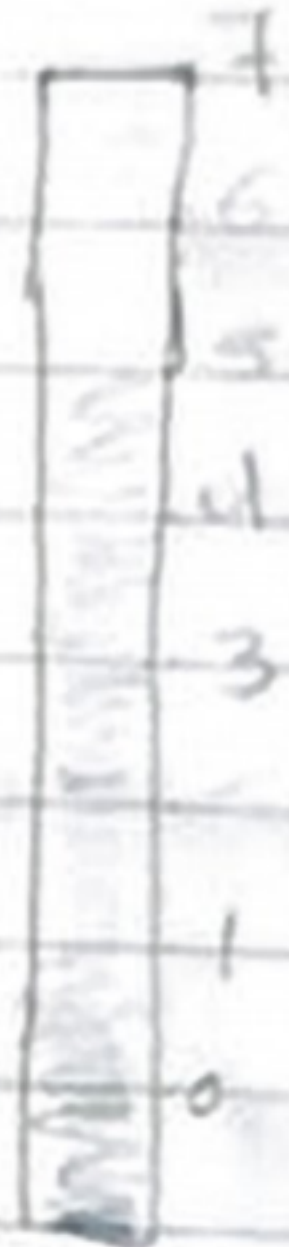
missing original image remaining graph (-1)

6/8

Q4
IP

MN $\rightarrow 4 \times 8 = 32$

r_k	n_k	$P_r(r_k)$	CDF	S_k	
0	1	$\frac{1}{32}$	$\frac{1}{32}$	0.21875	$\rightarrow 0$
1	4	$\frac{4}{32}$	$\frac{5}{32}$	1.09375	$\rightarrow 1$
2	8	$\frac{8}{32}$	$\frac{13}{32}$	2.84	$\rightarrow 3$
3	5	$\frac{5}{32}$	$\frac{18}{32}$	3.9	$\rightarrow 4$
4	2	$\frac{2}{32}$	$\frac{20}{32}$	4.375	$\rightarrow 4$
5	2	$\frac{2}{32}$	$\frac{22}{32}$	4.8	$\rightarrow 5$
6	5	$\frac{5}{32}$	$\frac{27}{32}$	5.9	$\rightarrow 6$
7	5	$\frac{5}{32}$	1	7	$\rightarrow 7$



Largest

r_q	n_q	$P_z(z_q)$	CDF	N_q	
0	0	0	0	0	$\rightarrow 0$
1	0	0	0	0	$\rightarrow 0$
2	4	$\frac{4}{32}$	$\frac{4}{32}$	0.875	$\rightarrow 1$
3	4	$\frac{4}{32}$	$\frac{8}{32}$	1.75	$\rightarrow 2$
4	4	$\frac{4}{32}$	$\frac{12}{32}$	2.625	$\rightarrow 3$
5	4	$\frac{4}{32}$	$\frac{16}{32}$	3.5	$\rightarrow 4$
6	5	$\frac{5}{32}$	$\frac{21}{32}$	4.16	$\rightarrow 5$
7	11	$\frac{11}{32}$	1	7	$\rightarrow 7$

Mapping

u	v
0	0 ✓
1	2 ✓
2	X
3	X
4	5 ✓
5	6 ✓
6	6 ✓
7	7 ✓

-0.5

Where is image or even Comment? (-1)

$$\frac{3.5}{5}$$